



Matteo Martinelli Phd

09/10/1991 | EMEA (Hybrid, Remote)

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PROFESSIONAL PROFILE

Management Engineer & Automation Specialist | Phd in Industrial Innovation Engineering

5+ Years of experience in Engineering & Digital Industries

Skills: Automation Engineering | Project Management | Lead Generation | Account Management

Tools: Programming skills & Code Design (Java, Python, Javascript), **Data Analysis** (Numpy, Pandas, Matplotlib), **Data & Systems** (SQL/NoSQL, Windows, Linux, macOS), **Simulations** (AnyLogic, PuLP), **CRM** (Twenty, Salesforce)

Certificazioni: Ph.D. in **Industrial Innovation Engineering** (2024), ACAI TAILOR Summer School (AI-themes, 2022), MSc in **Management Engineering** (2019), BSc in Management Engineering (2015).

Lingue: Italian (native language), English (C1, fluent), German (A2, base).

PROFESSIONAL EXPERIENCES

Senior Automation Engineer - Post Doctoral Research – UNIMORE

Reggio Emilia, Italy

Project Management | Lead Generation | Account Management

11/2024 – Ongoing

- **Generated 40+ industrial leads** by participating in 3 innovation events, 2 national fairs, and university-oriented events.
- **Coordinated a group of researcher analyst** activities, delivering **2 prototypes, 2 articles**, and the **restruction of an existing MicroFactory equipment**;
- Contributed to the **design of industrial use cases** in complex production and logistics environment based on the research group's skills and developed knowledge;
- Contributed to the **writing and delivery of 3+ industrial research projects**;
- Research Topics: **Industrial Digital Twins**, Cyber-Physical **Architectures**, system **augmentation**, **AI-relationships**, patterns of **specialization**, use cases investigation, industrial **design techniques**;
 - Contributed to the scientific research by publishing **10+ articles** in international journals as *Computers in Industry*, *Future Generation Computer Systems*, and related conferences;
 - Disseminated the research results by participating to **2+ international conferences**; contributed to the **organization of an international Special Issue** (Future Generation Computer Systems journal, 2026 - ongoing) and an **international scientific workshop in Las Vegas** (Industrial Digital twins and Emerging Technologies - IDETwin, Jan 2026); 🇺🇸

AI, Automation & Software Development (Professional Coach) – Freelance

Reggio Emilia, Italy

Professional Training | Live Coaching | Public speaking

11/2024 – Ongoing

- Delivered **530+ coaching hours** to **more than 180 students** from both **industry** and **specialization programs for professionals**.
- **Prepared more 5+ courses** with theoretical and practical material, setup and delivery of proficiency tests, **evaluation** of participants, coordination with training institutions;
- Main topics: Programming logics and **design tools**, **Python**, **Java** and **OOP**, data analysis with **Python**, **Docker** and containerization, principles of **cloud computing**, **AI** and **ML** principles, **Git** and GitHub usage.

Industrial Automation Expert - Ph.D. Candidate - UNIMORE

Reggio Emilia, Italy

Industrial Research | Scientific Writing | Public Speaking

12/2020 – 11/2024

- Contributed to the scientific research by studying **eXplainable AI** (xAI) models and their applicability in **industrial scenarios**, the role of **Digital Twins** in **industrial contexts**, synergies between **AI** and **Digital Twins**, publishing 5 scientific articles;
- Disseminated the research results by participating in 3 scientific international workshops.

Industrial Continuous Improvement & Safety Specialist - Ognibene Power

Reggio Emilia, Italy

Lean Manufacturing | SMED | Audit Presentation | Team Management

06/2019 – 12/2020

- Contributed to the **monitoring and improvement of industrial processes** in assigned areas, through planned projects, identified opportunities and stakeholders engagement.
- Improvement activities involved SMED, WO, EWO-SEWO, Micro-Areas, Tag Management and other Lean manufacturing tools.
- I **successfully completed the 2020 audit coordinating a team of 4 people**, guaranteeing production targets and safety standards and **saving a total of 128 h/year in setups**;

Master's Degree Internship - Industrial Project Lead - Dana Motion System

Reggio Emilia, Italy

Lean Manufacturing | Assembly line | Industrial AI

01/2019 – 06/2019

- I **re-designed an existing assembly line** of a family of mechanical reducers obtaining a **productivity raise from 46% to 81% completed on schedule** over 5 months.
- I developed an **AI-based Poka-Yoke Proof of Concept** to avoid frequent assembly errors, **classifying correctly more than 90% of correct classes**. Results have been published in a scientific journal.

EDUCATION & TRAINING

Guest Researcher at Universität Stuttgart



Stuttgart, Germany

Hosted at Institut für Steuerungstechnik der Werkzeugmaschinen (ISW)

10/2022 - 04/2023

- Topics: **Digital Twins** for **Industry 4.0** and industrial composability challenges.

TAILOR Summer School 2022 – Universitat Politècnica de Catalunya (UPC)



Barcelona, Spain

Advanced Course on AI

06/2022

- Topics: Challenges and research on **explainable** and **trustworthy AI**.
- Goals: awarded for **best research presentation**.

Master's Degree – University of Modena and Reggio Emilia (UNIMORE)

Reggio Emilia, Italy

Management Engineering

2011 – 2015

- Thesis: *"An assembly line re-design: the Dana Motion System case."*

Master's Degree – AGH-University of Science and Technology



Krakow, Poland

Management Engineering, Erasmus+ Program

2011 – 2015

Bachelor's Degree – University of Modena and Reggio Emilia (UNIMORE)

Reggio Emilia, Italy

Management Engineering

2011 – 2015

- Thesis: *"Implementation of a reading and translation system of DXF files into two axis PLC instructions"*, applied in the packaging industry.